```matlab
% run_alldredge_disagg_sweep.m
%
% Sweep disagg coefficient C0 across Parker-to-Alldredge range.
%
% Background:
%   Current model uses Parker et al. (1972): C0 = 9.39e-6 m
%   (derived from wastewater activated sludge).
%   Adrian suggests Alldredge marine-snow values which are larger
%   (weaker disagg per unit turbulence) because marine snow is
%   less dense and more fragile than wastewater flocs.
%
%   D_max = C0 * eps^(-1/4)
%
%   Values tested (all in metres):
%     9.39e-6   Parker original
%     1.88e-5   Parker x2
%     4.70e-5   Parker x5  <- current best from Da sweep
%     9.39e-5   Parker x10
%     2.35e-4   Parker x25
%     4.70e-4   Parker x50 (upper Alldredge-scale estimate)
%
% Fixed: alpha = 0.10, r0 = 0 (best from 2D grid, June 12)

script_dir = fileparts(mfilename('fullpath'));
addpath(script_dir);
addpath(fullfile(script_dir, '..', '..', 'src'));

mat_path = fullfile(script_dir, '..', '..', 'data', 'NA', ...
    'Turbulance', 'keps_for_dave.mat');
uvp_file = fullfile(script_dir, '..', '..', 'data', 'NA', 'uvp', 'raw', ...
    '745f00117e_EXPORTS-EXPORTSNA_UVP5-
ParticulateLevel2_differential_survey_20210504-20210529_R1.sb');

% ----------------------------------------------------------------
% 1. Shared setup
% ----------------------------------------------------------------
col_grid  = ColumnGrid(1000, 20);
keps_day  = load_keps_daily(mat_path, col_grid.z_centers);
prof      = load_keps(mat_path, col_grid.z_centers);

k_bc      = 2;
k_compare = 3:10;
z_compare = col_grid.z_centers(k_compare);

dt             = 0.25;
steps_per_day = round(1 / dt);
spinup_tol     = 0.01;
max_cycles     = 50;

Da_parker = 9.39e-6;   % Parker base [m]

cfg_base = SimulationConfig();
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cfg_base.n_sections       = 30;
cfg_base.sinking_law      = 'kriest_8';
cfg_base.disagg_mode      = 'operator_split';
cfg_base.disagg_dmax_cm   = 1.0;
cfg_base.disagg_dmax_A    = Da_parker * 5;    % starting point
cfg_base.enable_coag      = true;
cfg_base.enable_disagg    = true;
cfg_base.enable_zoo       = true;
cfg_base.enable_microbe   = false;
cfg_base.enable_mining    = true;
cfg_base.alpha            = 0.10;    % best from 2D grid
cfg_base.microbe_r0       = 0.0;
cfg_base.surface_pp_mu    = 0.0;
cfg_base.r_to_rg          = 1.6;
cfg_base.zoo_c            = 0.025;
cfg_base.zoo_s            = 1.3e-5;
cfg_base.zoo_p            = 0.5;
cfg_base.zoo_ic           = 7;
cfg_base.mining_s         = 1.3e-5;
cfg_base.fp_alpha_cross   = 0.5;
cfg_base.validate();

n_sec = cfg_base.n_sections;

% ---------------------------------------------------------------
% 2. BC at 100 m
% ---------------------------------------------------------------
bc = get_daily_bc_at_depth(uvp_file, cfg_base, col_grid, 100, k_compare);
phi_bc_daily    = bc.phi_bc_daily;
n_days          = bc.n_days;
id_model_best   = bc.id_model_best;
phi_uvp_cmp     = bc.phi_uvp_cmp;
mask_uvp_model = bc.mask_uvp_model;
fprintf('Best cast day: %d\n', bc.best_date);

% ---------------------------------------------------------------
% 3. C0 sweep
% ---------------------------------------------------------------
% Parker x multipliers: 1, 2, 5 (current), 10, 25, 50
c0_vals  = Da_parker * [1, 2, 5, 10, 25, 50];
c0_names = {'Parker x1', 'Parker x2', 'Parker x5 (current)', ...
             'Parker x10', 'Parker x25', 'Parker x50'};
n_cases  = numel(c0_vals);

ratio_table = NaN(numel(k_compare), n_cases);
scores      = NaN(1, n_cases);
Y_snaps     = cell(n_cases, 1);

for ic = 1:n_cases
    fprintf('\n=== %s  (C0 = %.2e m) ===\n', c0_names{ic}, c0_vals(ic));

    cfg = cfg_base;
    cfg.disagg_dmax_A = c0_vals(ic);
    cfg.validate();
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    sim = ColumnSimulation(cfg, col_grid, prof);
    Y   = zeros(col_grid.n_z, n_sec);
    Yfp = zeros(col_grid.n_z, n_sec);

    % spinup
    for icyc = 1:max_cycles
        phi_before = mean(sum(Y + Yfp, 2));
        for i_day = 1:n_days
            sim.rhs.profile.eps = keps_day.eps(:, i_day);
            for i_step = 1:steps_per_day
                Y(k_bc, :) = phi_bc_daily(i_day, :);
                [Y, Yfp] = sim.rhs.stepY(Y, dt, Yfp);
                Y(k_bc, :) = phi_bc_daily(i_day, :);
            end
        end
        phi_after  = mean(sum(Y + Yfp, 2));
        rel_change = abs(phi_after - phi_before) / max(phi_before, 1e-20);
        if rel_change < spinup_tol
            fprintf('  Converged at cycle %d\n', icyc);
            break;
        end
    end

    % final run
    Y   = zeros(col_grid.n_z, n_sec);
    Yfp = zeros(col_grid.n_z, n_sec);
    for i_day = 1:n_days
        sim.rhs.profile.eps = keps_day.eps(:, i_day);
        for i_step = 1:steps_per_day
            Y(k_bc, :) = phi_bc_daily(i_day, :);
            [Y, Yfp] = sim.rhs.stepY(Y, dt, Yfp);
            Y(k_bc, :) = phi_bc_daily(i_day, :);
        end
        if i_day == id_model_best
            Y_snaps{ic} = Y + Yfp;
        end
    end

    phi_mod = sum(Y_snaps{ic}(k_compare, mask_uvp_model), 2);
    ratio_table(:, ic) = phi_mod ./ max(phi_uvp_cmp, 1e-30);
    scores(ic) = mean((ratio_table(:, ic) - 1).^2);
end

% -------------------------------------------------------------
% 4. Print results
% -------------------------------------------------------------
fprintf('\n--- Alldredge sweep: model / UVP  (alpha=0.10, r0=0) ---\n');
fprintf('%-10s', 'Depth(m)');
for ic = 1:n_cases
    fprintf('  %8s', sprintf('x%d', round(c0_vals(ic)/Da_parker)));
end
fprintf('\n%s\n', repmat('-', 1, 10 + n_cases*10));
for i = 1:numel(k_compare)
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    fprintf('%-10.0f', z_compare(i));
    for ic = 1:n_cases
        fprintf('  %8.2f', ratio_table(i, ic));
    end
    fprintf('\n');
end
fprintf('\nScore (lower=better):\n');
for ic = 1:n_cases
    fprintf('  %-25s  %.4f\n', c0_names{ic}, scores(ic));
end
[~, ibest] = min(scores);
fprintf('\nBest: %s\n', c0_names{ibest});

% ---------------------------------------------------------------
% 5. Figure
% ---------------------------------------------------------------
fig_dir = fullfile(script_dir, '..', '..', 'docs', 'figures');
if ~exist(fig_dir, 'dir'), mkdir(fig_dir); end

colors = lines(n_cases);
figure('Units','centimeters','Position',[2 2 11 13]);
hold on;
for ic = 1:n_cases
    plot(ratio_table(:, ic), z_compare, '-o', ...
        'Color', colors(ic,:), 'LineWidth', 1.5, ...
        'DisplayName', c0_names{ic});
end
xline(1.0, 'k--');
set(gca, 'YDir', 'reverse');
xlabel('model / UVP');
ylabel('Depth (m)');
ylim([100 500]);
xlim([0 2.5]);
legend('location', 'northeast', 'FontSize', 7);
title('Alldredge C_0 sweep: \alpha=0.10, r_0=0');

saveas(gcf, fullfile(fig_dir, 'alldredge_disagg_sweep.png'));
fprintf('\nSaved alldredge_disagg_sweep.png\n');
```

```
Best cast day: 20210522

=== Parker x1  (C0 = 9.39e-06 m) ===
  Converged at cycle 13

=== Parker x2  (C0 = 1.88e-05 m) ===
  Converged at cycle 12

=== Parker x5 (current)  (C0 = 4.70e-05 m) ===
  Converged at cycle 11

=== Parker x10  (C0 = 9.39e-05 m) ===
  Converged at cycle 10

=== Parker x25  (C0 = 2.35e-04 m) ===
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  Converged at cycle 9

=== Parker x50  (C0 = 4.69e-04 m) ===
  Converged at cycle 9

--- Alldredge sweep: model / UVP  (alpha=0.10, r0=0) ---
Depth(m)            x1         x2         x5         x10        x25        x50
----------------------------------------------------------------------
125              1.12       1.08       1.03       0.84       0.83       0.83
175              0.97       1.02       1.09       0.79       0.75       0.75
225              0.92       1.07       1.29       0.91       0.83       0.83
275              0.58       0.77       1.07       0.75       0.68       0.67
325              0.43       0.66       1.08       0.78       0.70       0.70
375              0.23       0.42       0.82       0.62       0.57       0.56
425              0.14       0.29       0.70       0.56       0.53       0.52
475              0.07       0.15       0.45       0.39       0.37       0.37

Score (lower=better):
  Parker x1                  0.3405
  Parker x2                  0.2165
  Parker x5 (current)        0.0659
  Parker x10                 0.1118
  Parker x25                 0.1394
  Parker x50                 0.1418

Best: Parker x5 (current)

Saved alldredge_disagg_sweep.png
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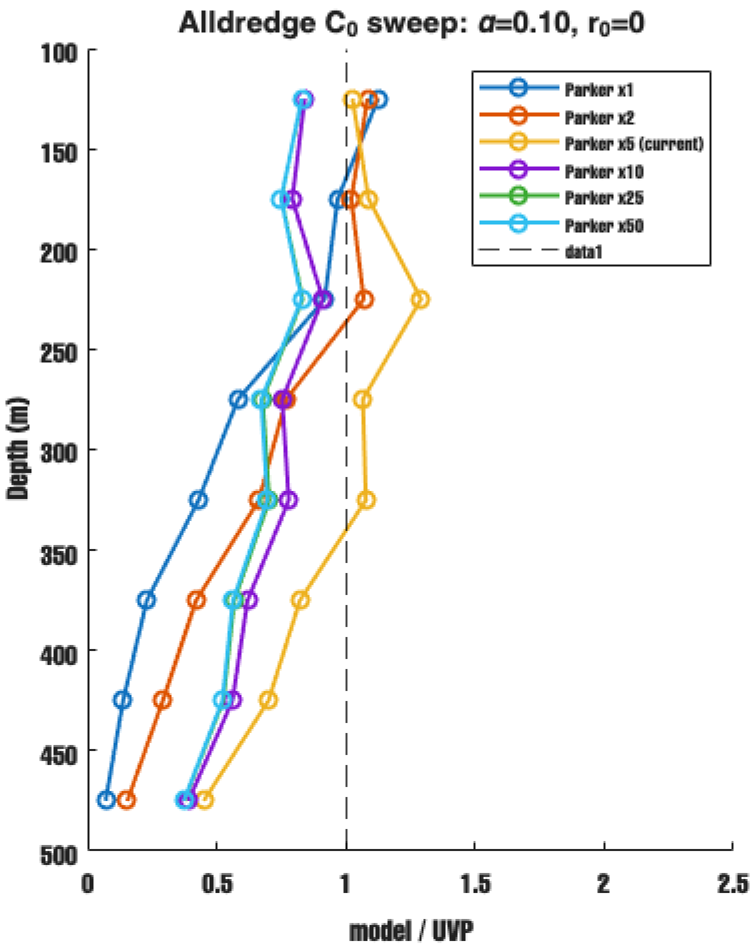

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